

How To Create a Microsoft 365 Hybrid Sync environment (On-Premises and M365 Cloud) And Syncing your On-Premises Users to the Cloud

Product Name: Microsoft Exchange, Azure Active Directory, Entra ID, Microsoft Entra Connect, Microsoft 365

Version: Entra Connect Sync 2.6.1.0 / Exchange 2019 CU15 /Windows Server 2022 DataCenter

Phase: Deployment

Date: 📅 Apr 7, 2026

Overview

Microsoft 365 (M365)

Microsoft 365

is a **cloud-powered subscription service** from Microsoft that bundles premium productivity apps with cloud services and advanced security features

- **Email → Outlook/Exchange**
- **File storage → OneDrive**
- **Collaboration → Teams, SharePoint**
- **Apps → Word, Excel, PowerPoint**
- **Security - Microsoft Defender**

In short:

👉 It helps you **work, communicate, and collaborate**.

Microsoft Entra (formerly Azure AD)

Microsoft Entra

A **cloud identity and access management platform** that controls **who can access what**.

Key features:

- User sign-in & authentication
- Single Sign-On (SSO)
- Multi-Factor Authentication (MFA)
- Conditional Access policies
- Identity protection

Microsoft 365 Exchange (Exchange 2019/Exchange Online)

Microsoft Exchange Online

Microsoft 365 Exchange is the **email and calendaring service** in Microsoft 365.

What it does:

-  **Email hosting** – Sends, receives, and stores business email
-  **Calendars** – Scheduling meetings, shared calendars
-  **Contacts** – Centralized address books (GAL)
-  **Sync across devices** – Outlook, mobile, web access
-  **Security & compliance** – Spam filtering, malware protection, retention policies

In simple terms:

👉 It's the system that **runs your company's email and scheduling**.

Product Objectives

["How To Create a Microsoft 365 Hybrid environment \(On-Premises and M365 Cloud\)"](#)

This document walks IT Professionals and Businesses through building a Microsoft Hybrid (On-Prem+Cloud) environment from scratch—installing Windows Server, configuring Active Directory, and setting up Exchange Email Services, Setting up

Exchange Online/Entra AD communication endpoints to the Internet and your Internal Business.

👉 This helps users avoid guesswork and follow a proven process.

2. Reduces errors and troubleshooting time

You include important checks (like DNS setup, prerequisites, and PowerShell commands).

👉 This helps users **avoid common mistakes** that could break things later.

3. Helps users achieve the end goal

The goal of your product/guide is to:

- Build an AD environment
- Deploy Exchange
- Create mailboxes and send email

How to Hybrid-join Microsoft Active Directory with Microsoft Entra ID 365

Hybrid Connect Entra Connect and Cloud Sync

<https://learn.microsoft.com/en-us/exchange/hybrid-deployment/deploy-hybrid>

<https://learn.microsoft.com/en-us/exchange/hybrid-deployment-prerequisites>

<https://learn.microsoft.com/en-us/microsoft-365/admin/setup/add-domain>

<https://learn.microsoft.com/en-us/microsoft-365/enterprise/urls-and-ip-address-ranges?view=o365-worldwide>

<https://learn.microsoft.com/en-us/microsoft-365/admin/get-help-with-domains/create-dns-records-at-any-dns-hosting-provider?view=o365-worldwide&tabs=domain-connect>

First we need to check that Pre-requisites are met -

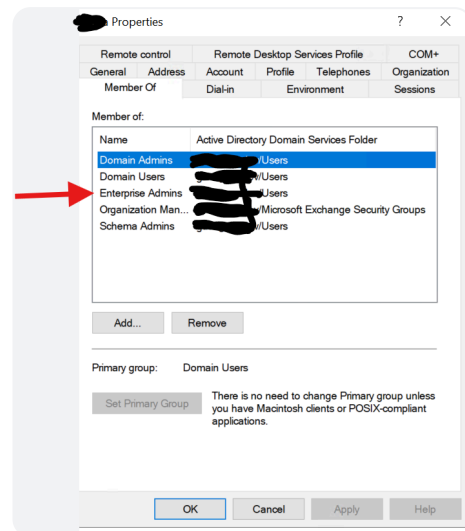
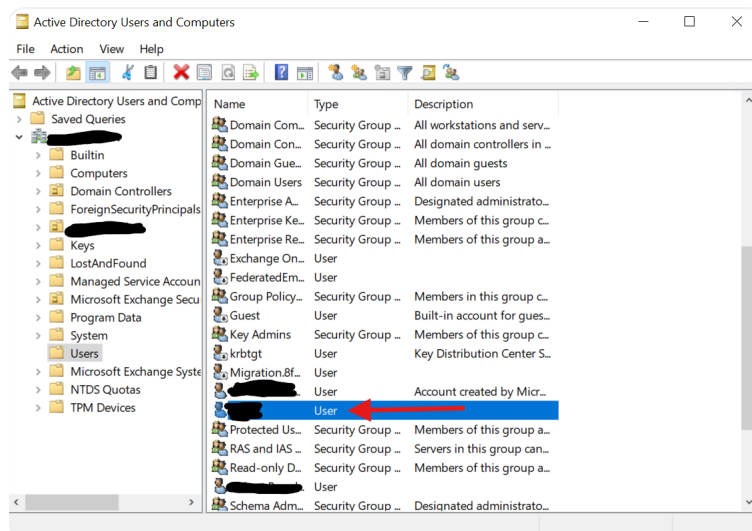
From your **On-Premise Server** - ensure you have a **User/Admin** assigned : **Enterprise Admin Role** credentials for on-prem Active Directory and a **Licensed Windows Server (2025,2022,2019) Standard or Datacenter Editions**

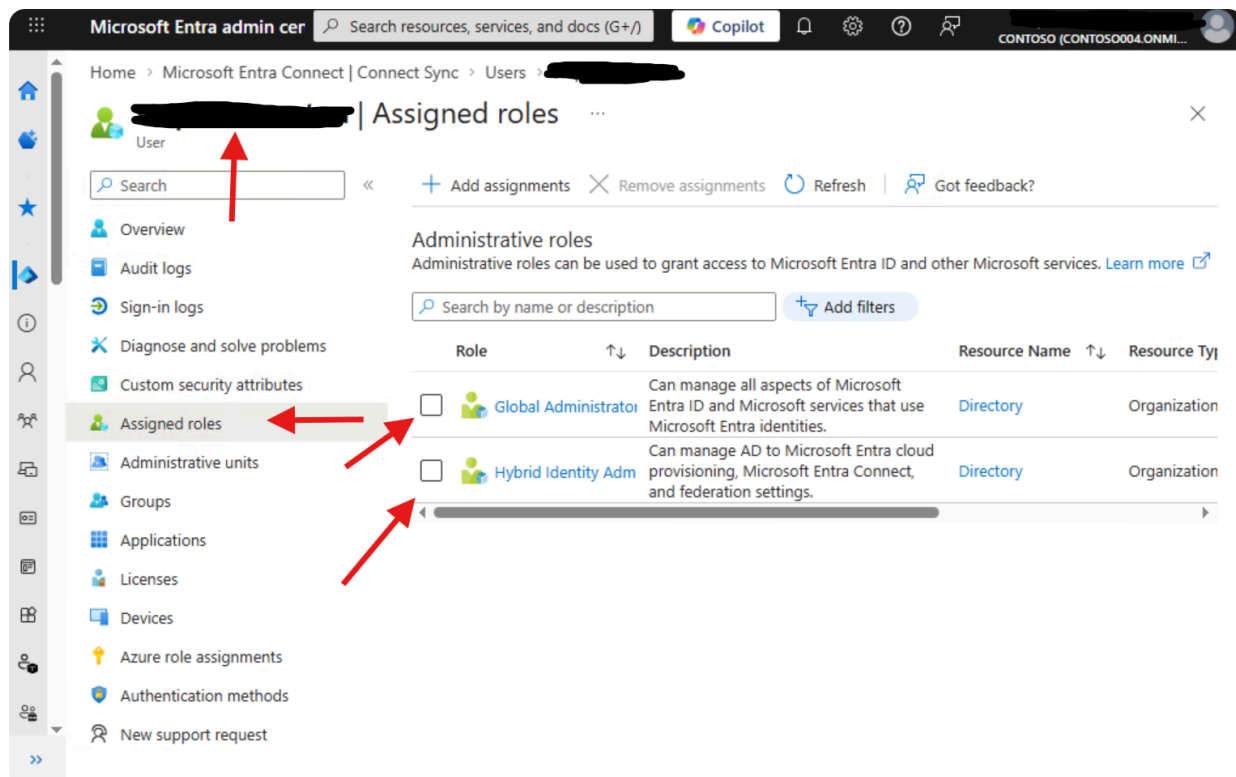
From your **M365 Cloud** - ensure you have a **User/Admin** with the **Hybrid Identity Administrator Role** for **Entra Identity** . (Images Shown Below)

Windows specifications

Edition Windows Server 2022 Datacenter
Version [REDACTED]
Installed on 4/1/2026
OS build 20348.1487

Copy





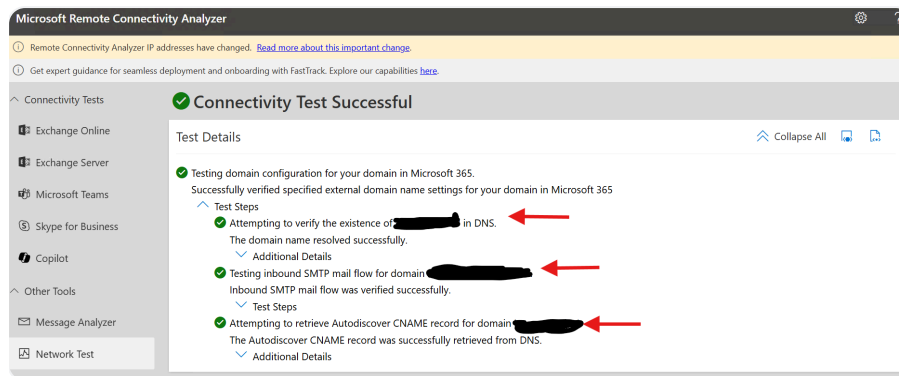
This URL Explains the Requirement for your Hybrid Deployment

<https://learn.microsoft.com/en-us/exchange/hybrid-deployment/deploy-hybrid>

- **Exchange 2016** and newer: At least one **Mailbox server**.
- **Active Directory synchronization: Deploy the Microsoft Entra Connect or cloud sync tool to enable Active Directory synchronization with your on-premises organization.**
- **Custom domains:** Register any custom domains you want to use in your hybrid deployment with Microsoft 365 or Office 365. You can do this by using the Microsoft 365 portal, or by optionally configuring **Active Directory Federation Services (AD FS)** in your on-premises organization.
- Make sure you have actively allow ports to communicate between **on-prem** and **M365 (Port 25 SMTP, 143/992 IMAP 80/443 HTTP/HTTPS, 587 SMB and 53 DNS)**
- **Active Directory synchronization between the on-premises organization and the cloud, which is performed every 30 minutes by a server running Microsoft Entra Connect, is a requirement for configuring a hybrid deployment.**
- And lastly, perform a check for your HCW readiness with Remote Connectivity Analyzer

The Microsoft Remote Connectivity Analyzer tool checks the external connectivity of your on-premises Exchange organization and makes sure that you're ready to configure your hybrid deployment.

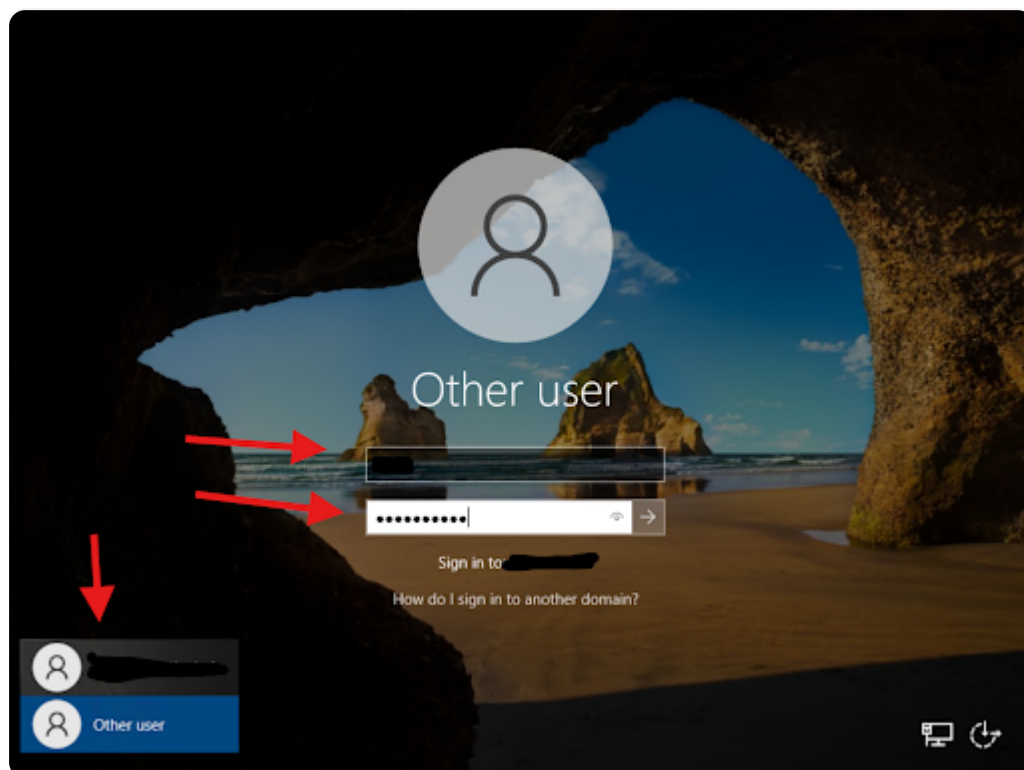
<https://testconnectivity.microsoft.com/>



Section 2:

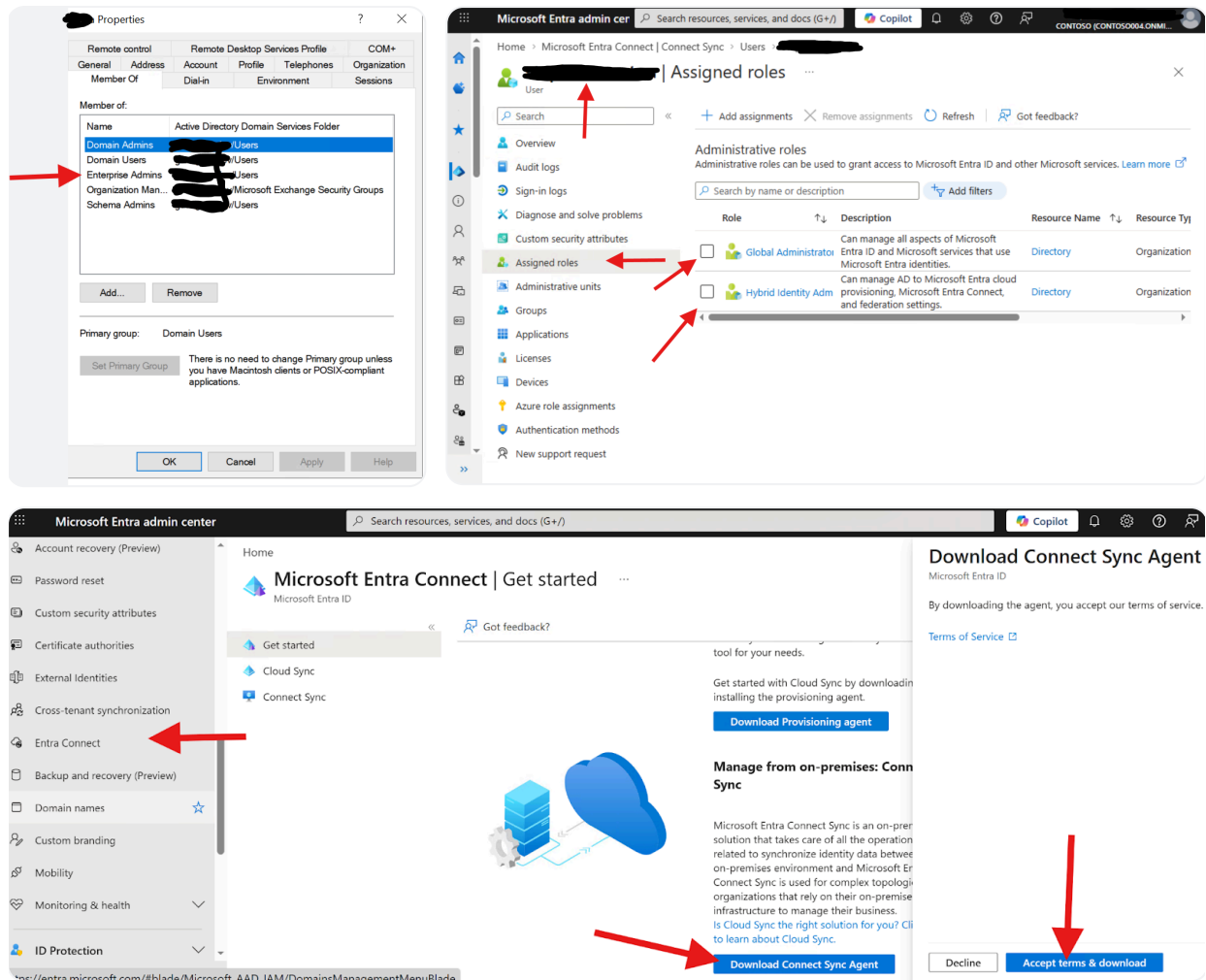
Installing Entra Connect for Azure AD Sync And Cloud Sync On-Premise

- First Sign-in to your **On-Prem Active Directory** With your "**Enterprise Admin**" or "**Domain Admin**" account

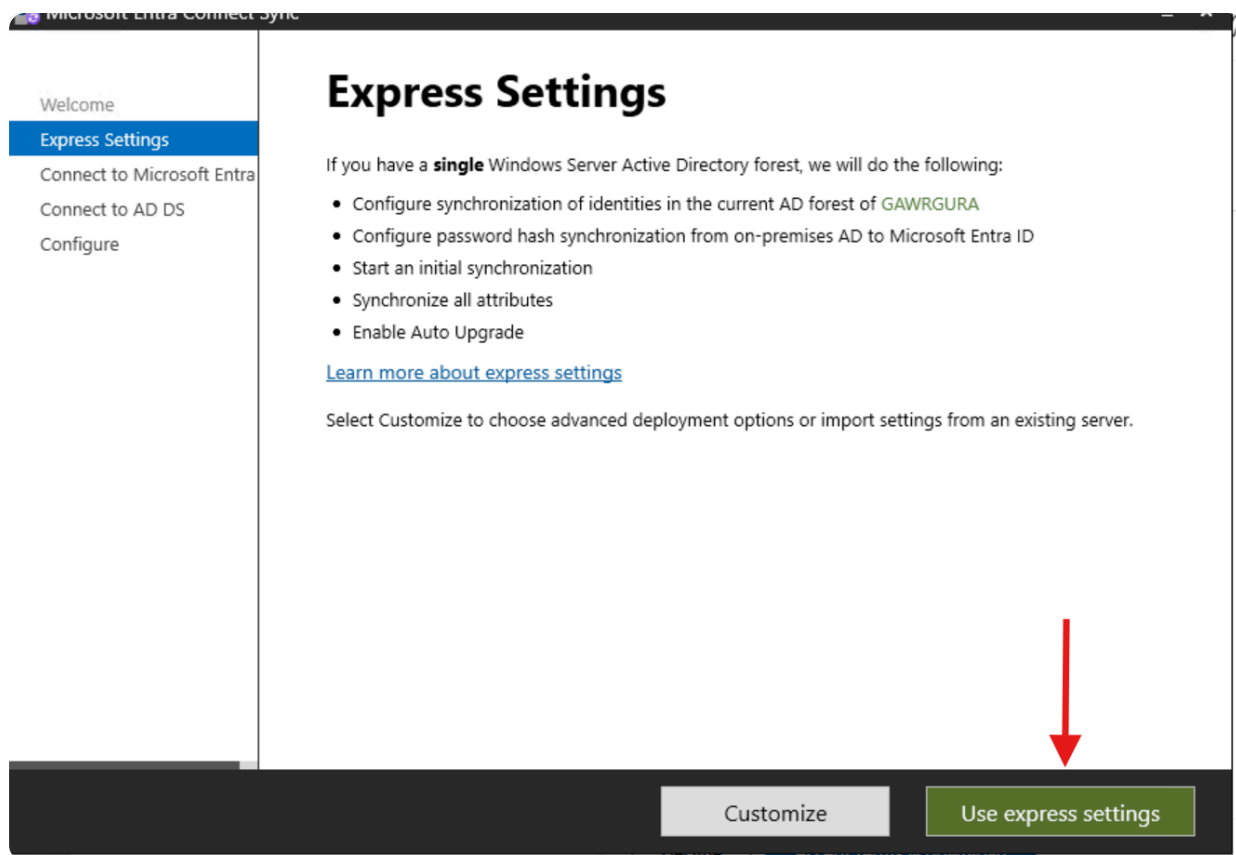


- Second go to your **Microsoft Entra admin Center** Sign-in with your **Global Administrator** account - Go to **Entra ID Tab**- Select "**Entra Connect**" for On-prem

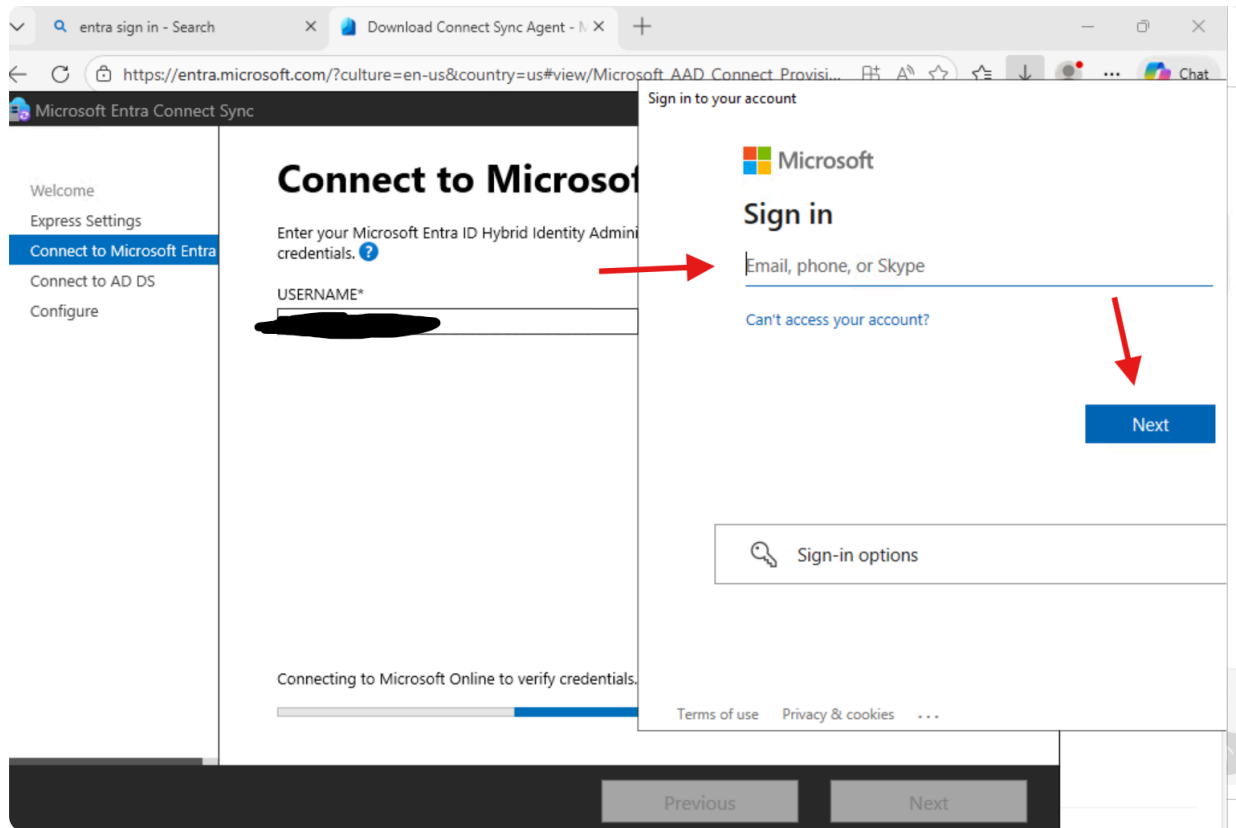
Syncing Click the “**Download Connect Sync Agent And Accept the Terms and Conditions.**



Since we Already Have an **Exchange** and **Active directory** we will Go through the “**Express Settings**” unless any Customization is needed. **IMPORTANT** - BE SURE YOU KNOW THE CREDENTIALS FOR YOUR ON-PREM ENTERPRISE ADMIN / AND YOUR CLOUD ADMIN (**Hybrid Identity Admin/Domain Name Admin**)



- **Connect Your M365 Entra Hybrid Identity Administrator to the "Username" Area Microsoft MFA will Pop-up for you to sign-in the Your Hybrid Admins Credentials.**



- Next, Enter your **AD DS Enterprise administrator** Credentials to Sync both **On-premises Exchange 2019** and **Microsoft Entra 365** Administrator Accounts.

Microsoft Entra Connect Sync

Welcome

Express Settings

Connect to Microsoft Entra

Connect to AD DS

Configure

Connect to AD DS

Enter the Active Directory Domain Services enterprise administrator credentials: ?

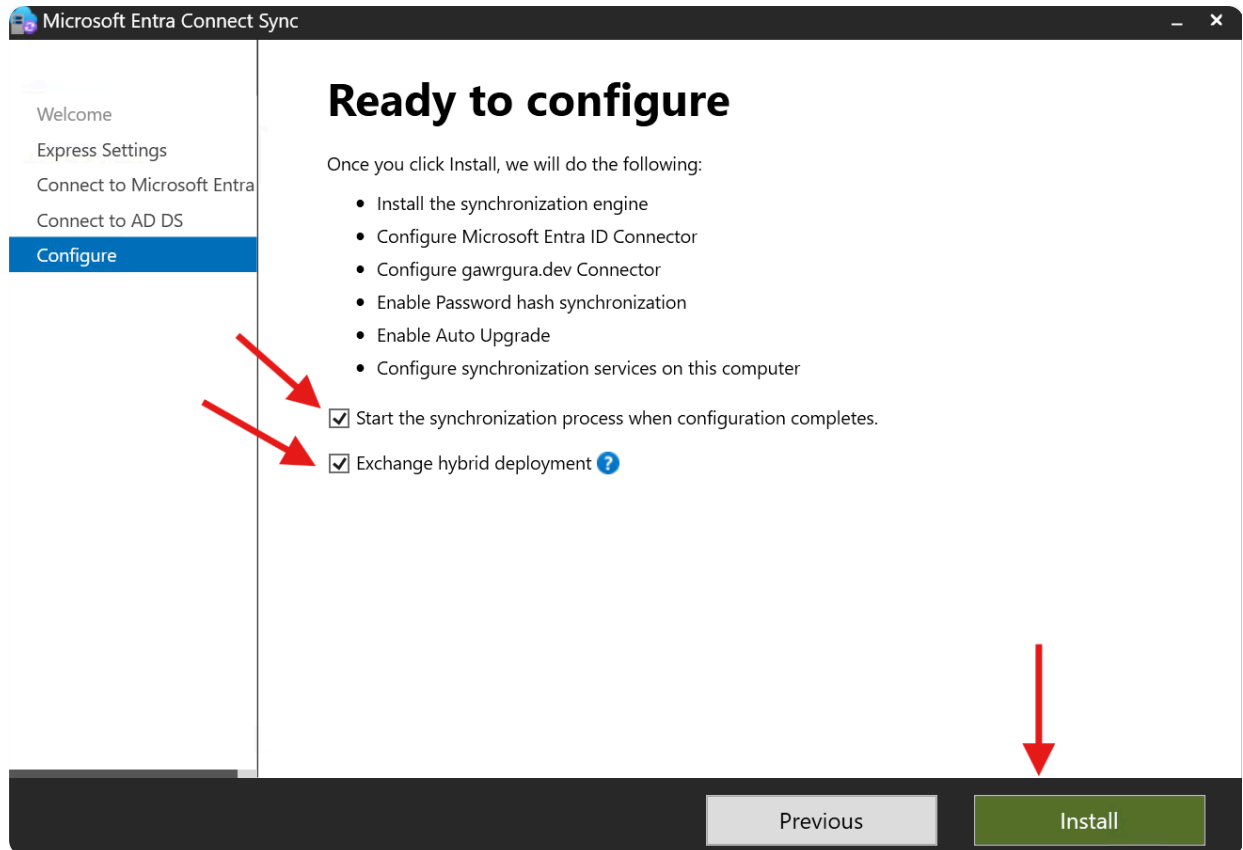
USERNAME

PASSWORD

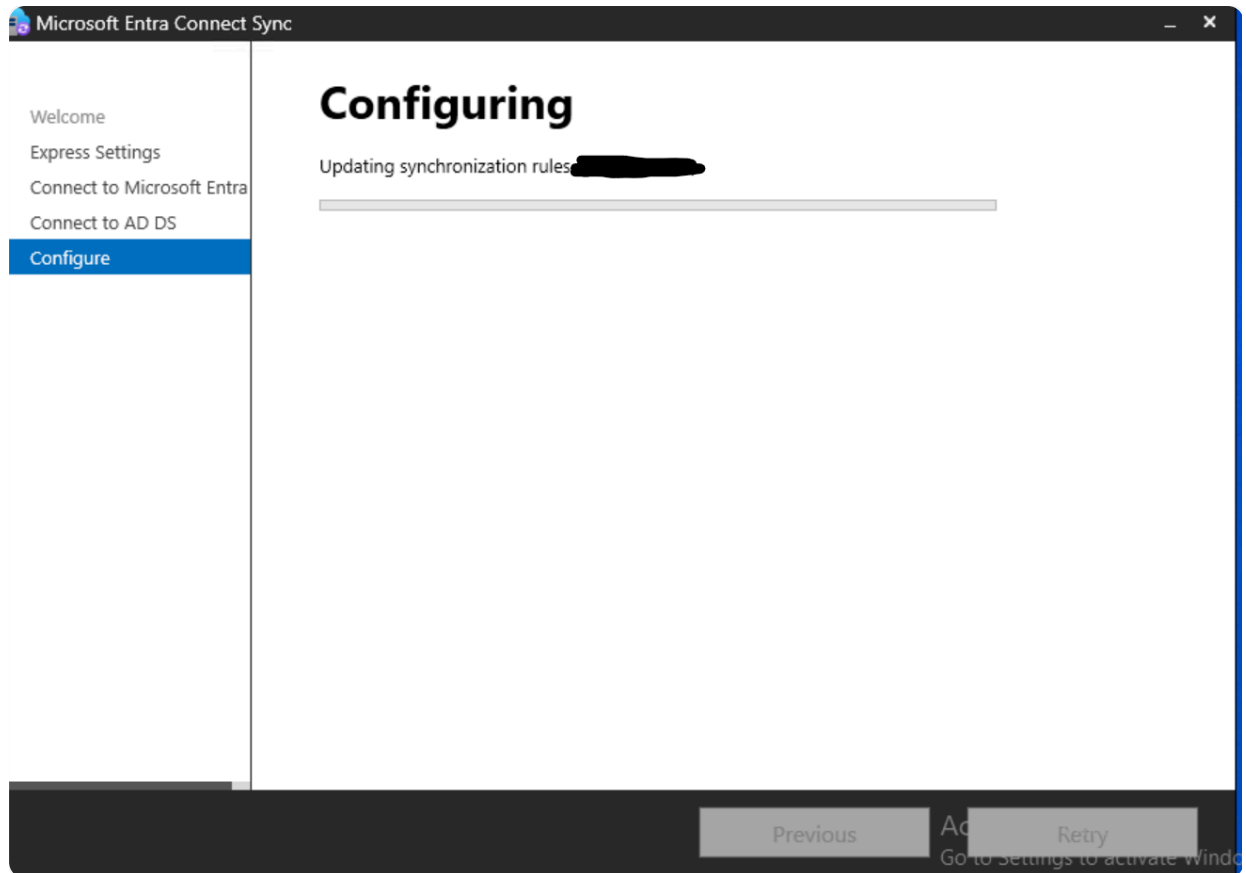
Previous Next

- **Once you have Reached the "Ready To Configure" Panel Be sure to checkmark Both "Start the Synchronization Process" And "Exchange Hybrid Deployment"**

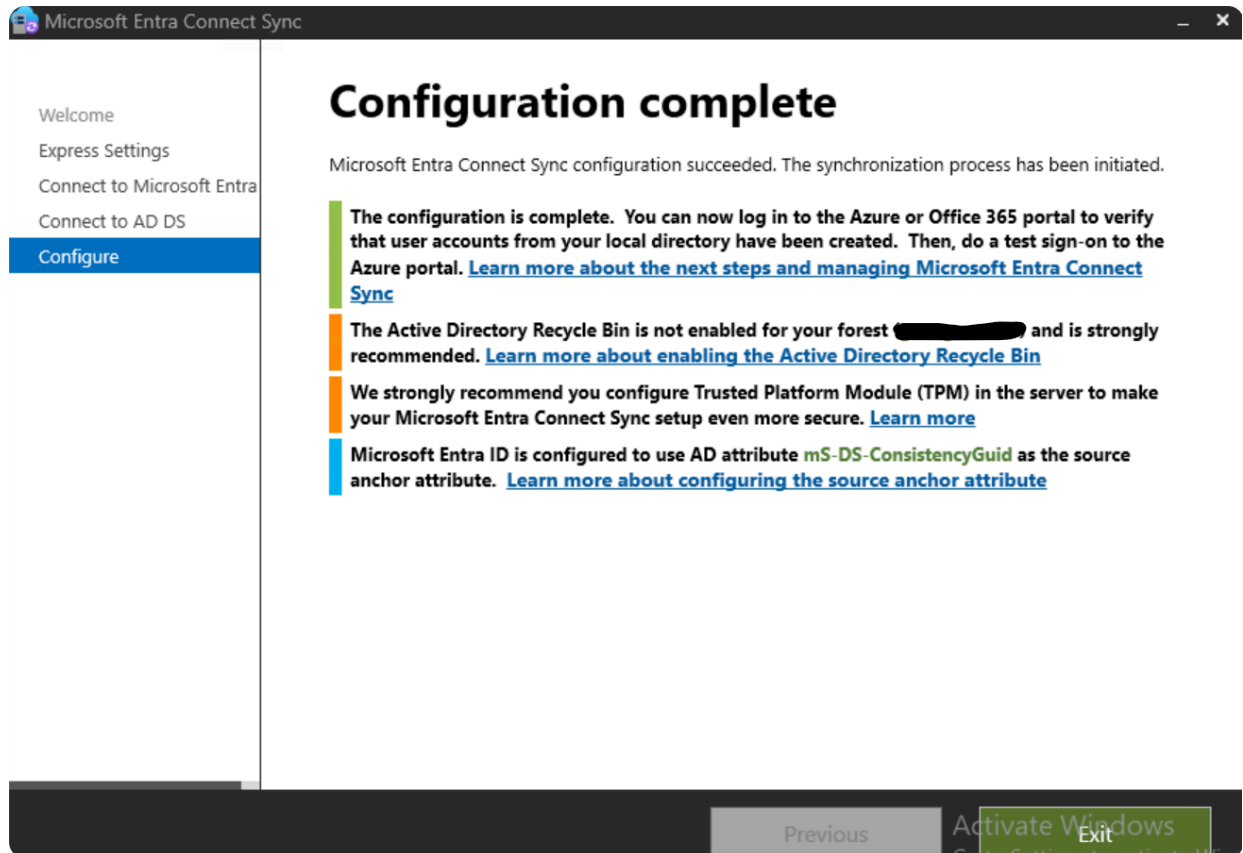
**Enables synchronization of specific Exchange attributes between on-premises Active Directory and Exchange Online (Office 365). It is crucial for co-existence, allowing shared address lists, calendar free/busy sharing, and proper mailbox management during or after migrations.*



Once you've Press "Install" wait for the **Entra Connect Sync** to Update Synchronization Rules/Data and prepare your **On-Premises** and **Microsoft 365 Cloud Tenant** to build a communication to sync **Attributes** and **User/Computer Identities**.

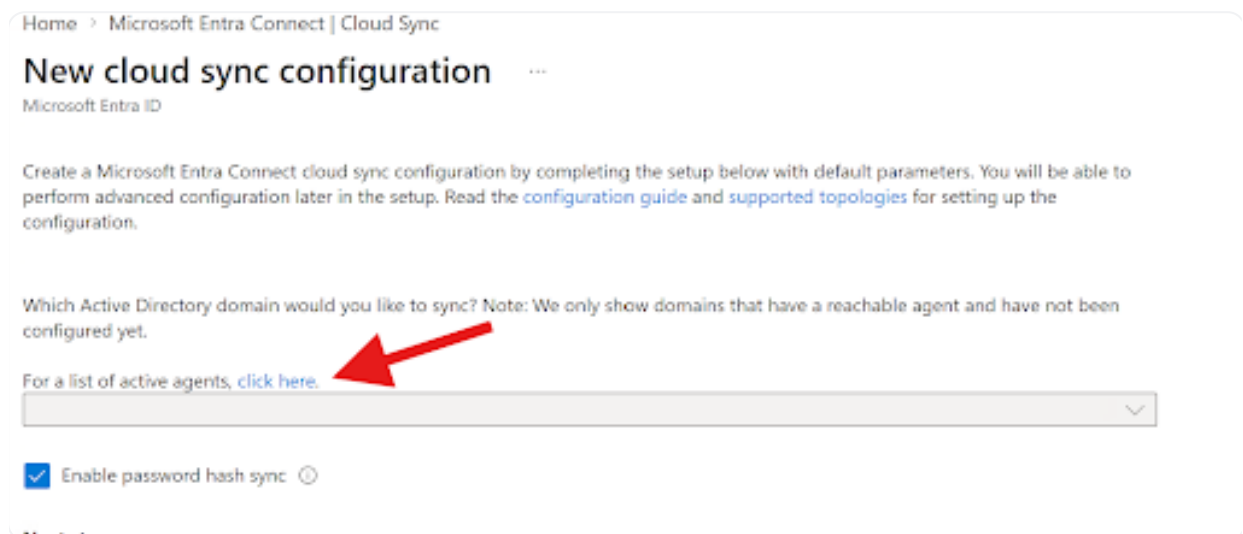


The Wizard has now created a communicated Hand-shake between your **Azure AD/**
Entra ID Objects, Identities, and Users/Computers.

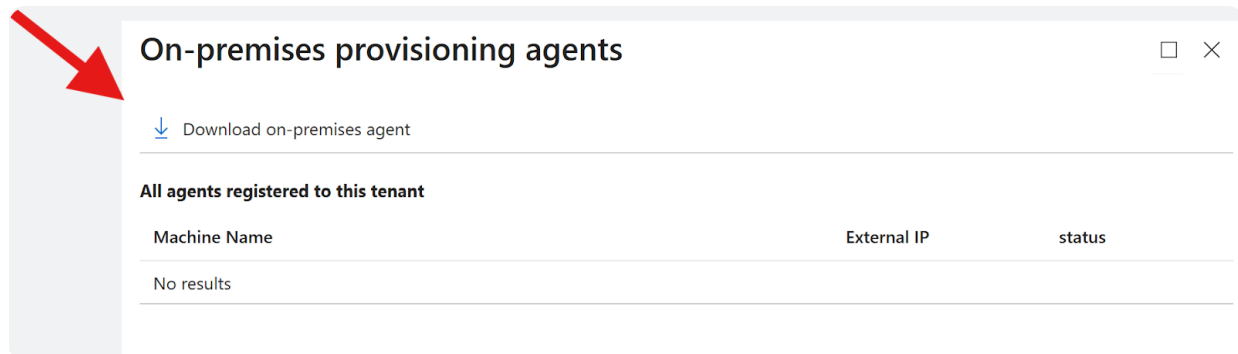


Section 3: Performing a Microsoft Entra Cloud Sync

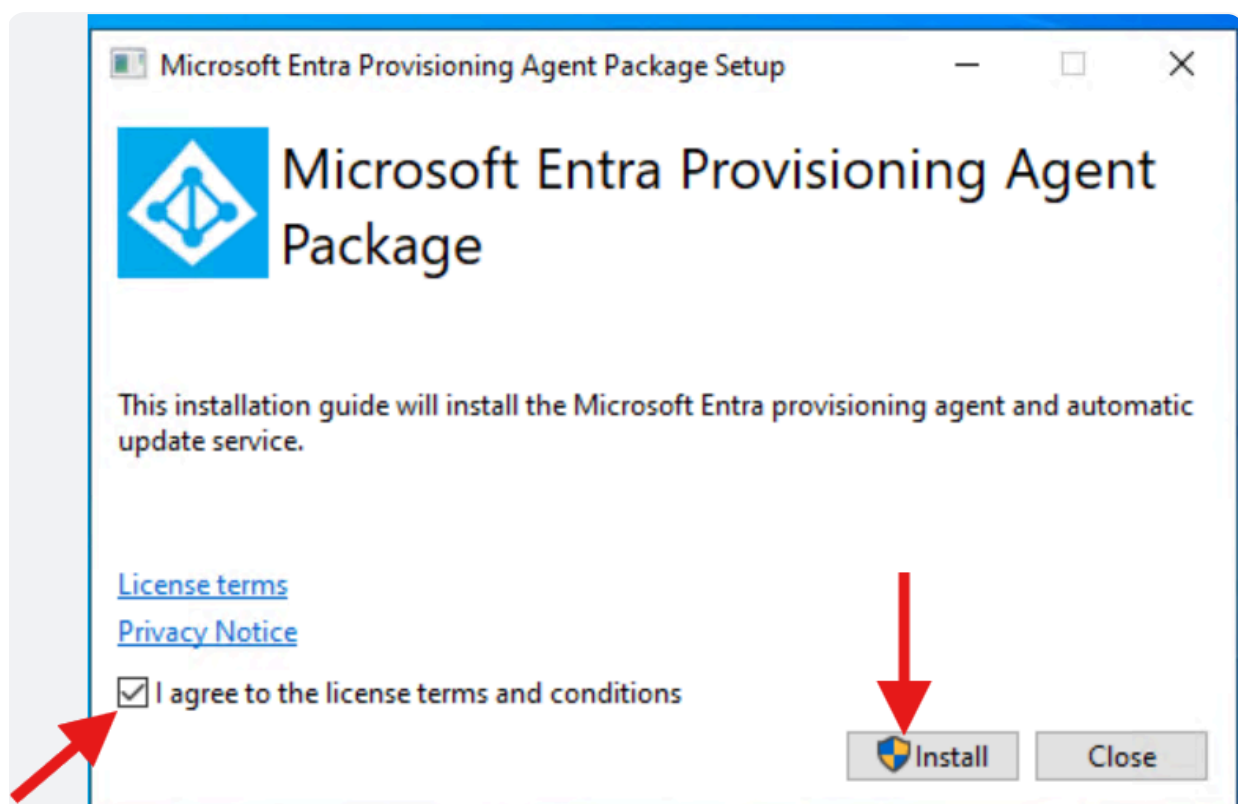
- First within "M365 Entra Admin Center" under "Entra connect" Select "Cloud Sync Configuration" Once you've come to the Drop-Down Bar you will read - For A List of active Agents select "Click Here"



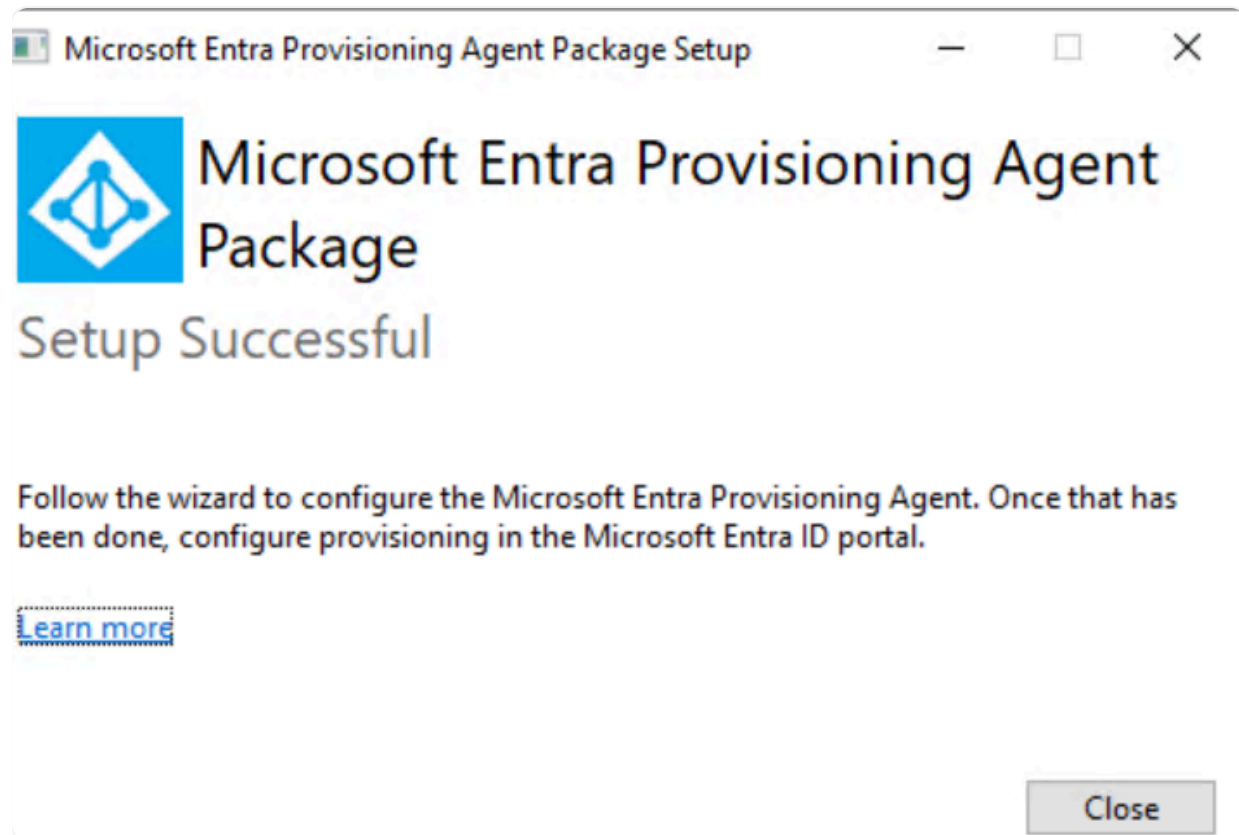
You Will then Get a Tab on the Right side of your Browser to “Download”



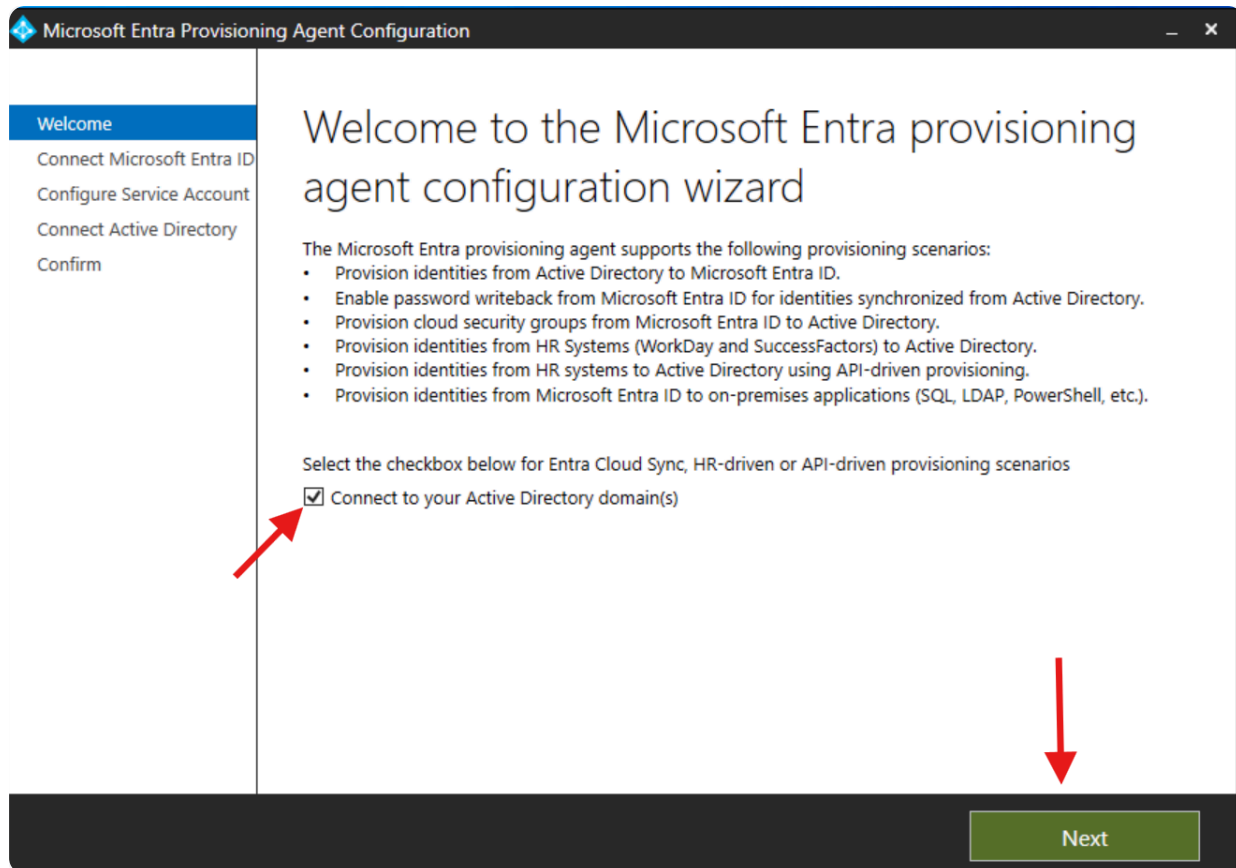
- Once downloaded to your machine “**Run As an Administrator**” This Pop-up will appear before you begin the Wizard Check the **Agree terms and Install with Admin Privileges**.



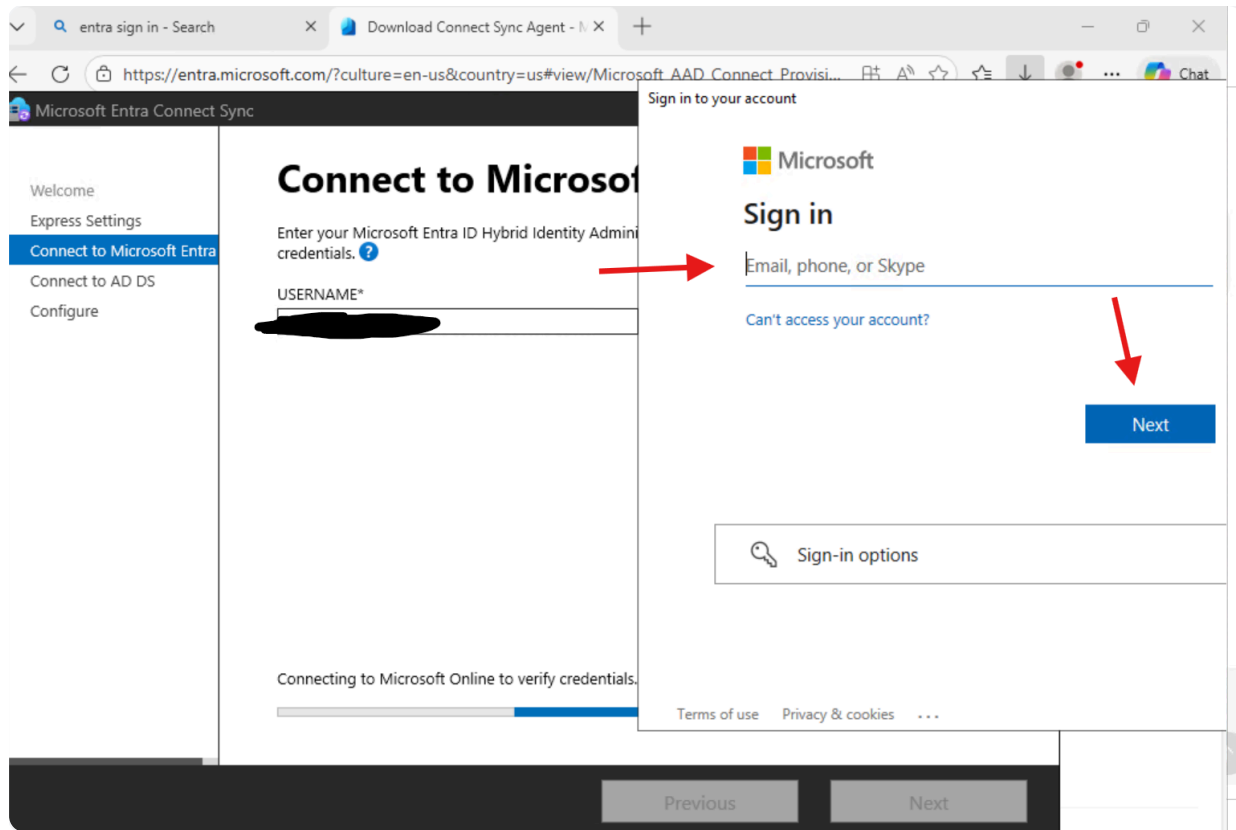
You will receive this pop-up as a Pre-check Click "**Close**" and begin the Wizard



- ✓ **Checkmark** this page to **connect your Active Directory Domain(s)** to prepare for Hybrid Setup. Click "**Next**"



Now connect the wizard to your Microsoft Entra ID Cloud Instance with your Cloud Admin Account - Then Click “**Authenticate**” Using your 2-MFA then click “**Next**”



Which gMSA do I choose? Custom or Standalone?

Welcome

Connect Microsoft Entra ID

Configure Service Account

Connect Active Directory

Confirm

Configure Service Account

Setup a group managed service account (gMSA) to manage your synchronization from Active Directory

☒ Create gMSA

☐ Use custom gMSA

Enter your domain admin credentials to setup the gMSA account for your Active Directory.

DOMAIN ADMIN USERNAME ?

Password

[Learn more about group managed service accounts \(gMSA\). Learn more.](#)

Previous Next

*In almost all modern Active Directory scenarios, a Group Managed Service Account (**gMSA**) is **better than a standalone** Managed Service Account (**sMSA**), primarily because it provides the same security benefits (automatic password management) but can be used across multiple servers.*

sMSA (Standalone): *Restricted to a single, specific computer account.*

gMSA: (Custom): *Designed for multiple servers, load balancers scenarios or server farms*

For Our Needs sMSA (standalone) will do what we need for our Lab Environment.

- Next We will grab the Connection Credentials from our on-premises Active Directory to provide the Tenants Last needed Information

Welcome

Connect Microsoft Entra ID

Configure Service Account

Connect Active Directory

Confirm

Connect Active Directory

Enter connection information for your on-premises Active Directory domains.

DOMAIN ?

Add Directory

CONFIGURED DOMAINS

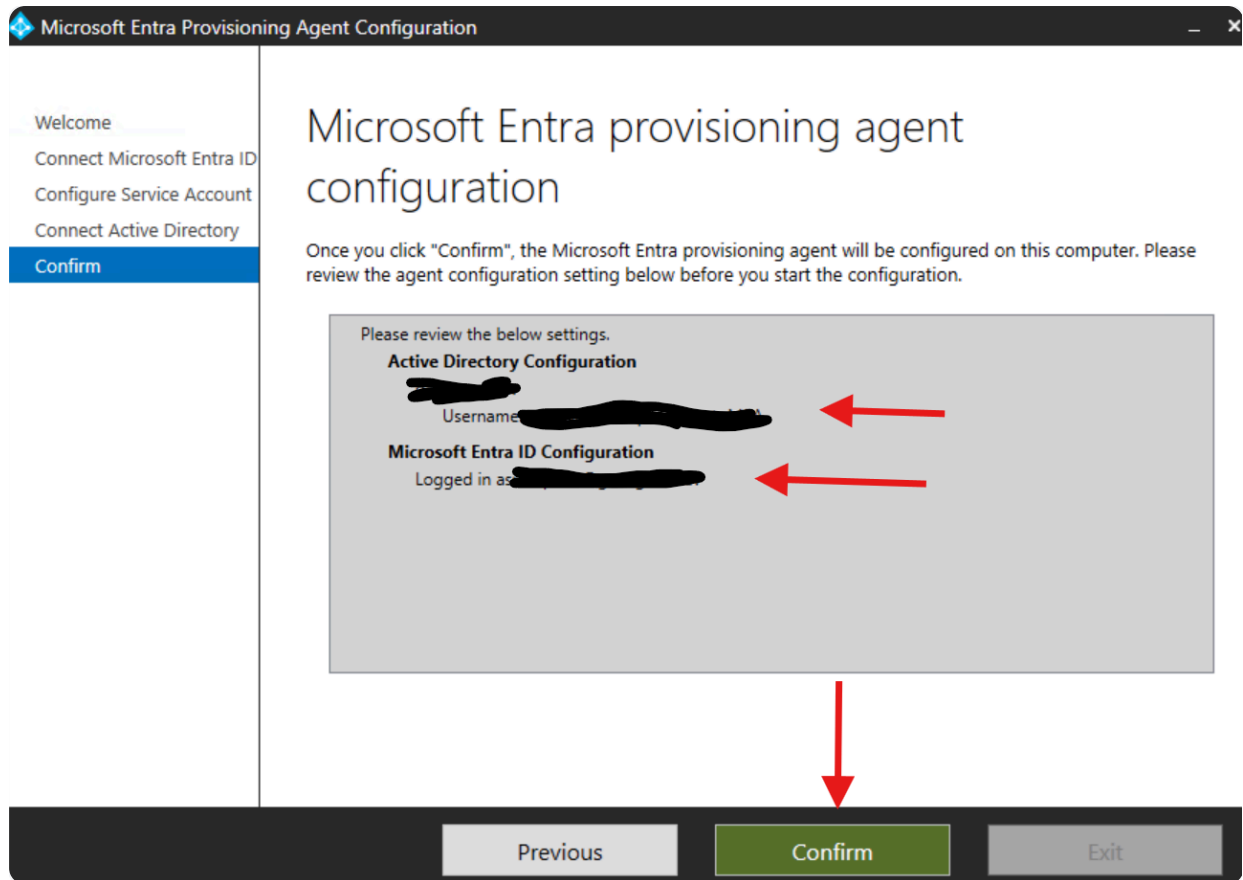
Change Credentials

Remove

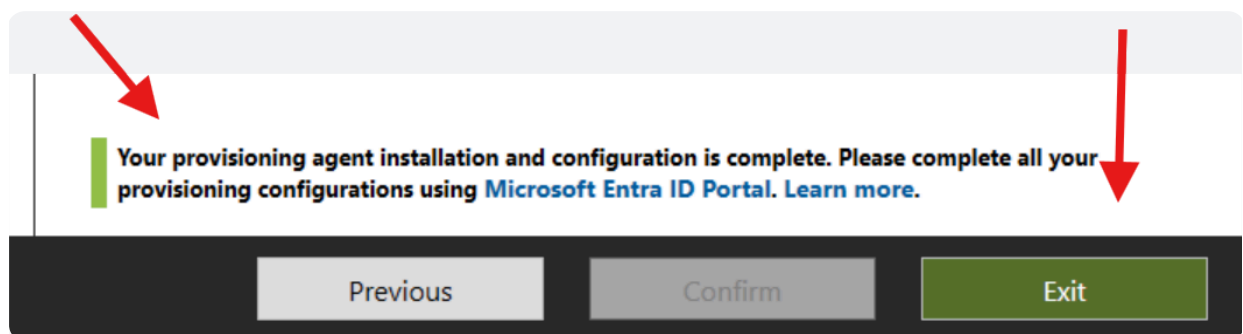
Previous

Next

Lastly Confirm the Correct admin Credentials have been added to do the Agent provisioning on your machine.

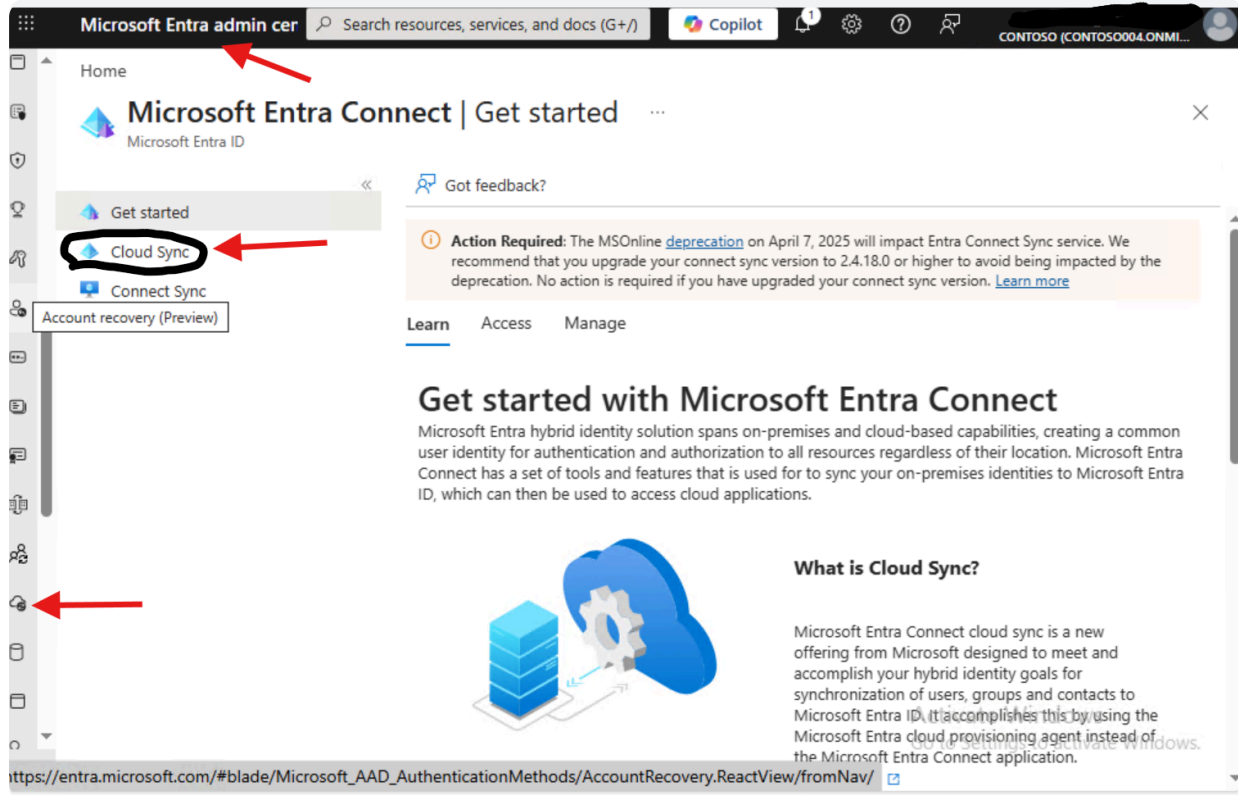


Now That Provisioning is done you can Press “Exit”



Now Return Back to your “M365 Entra admin Center”

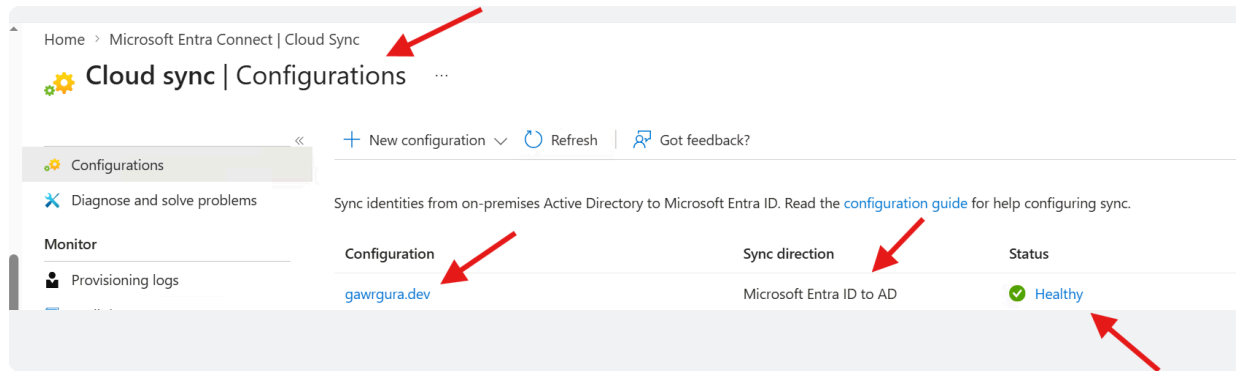
Go to “Entra Connect”, Then “Cloud Sync”



- Under "Cloud Sync | Configurations" we now see our first sync connection to make a "Hybrid Configuration "

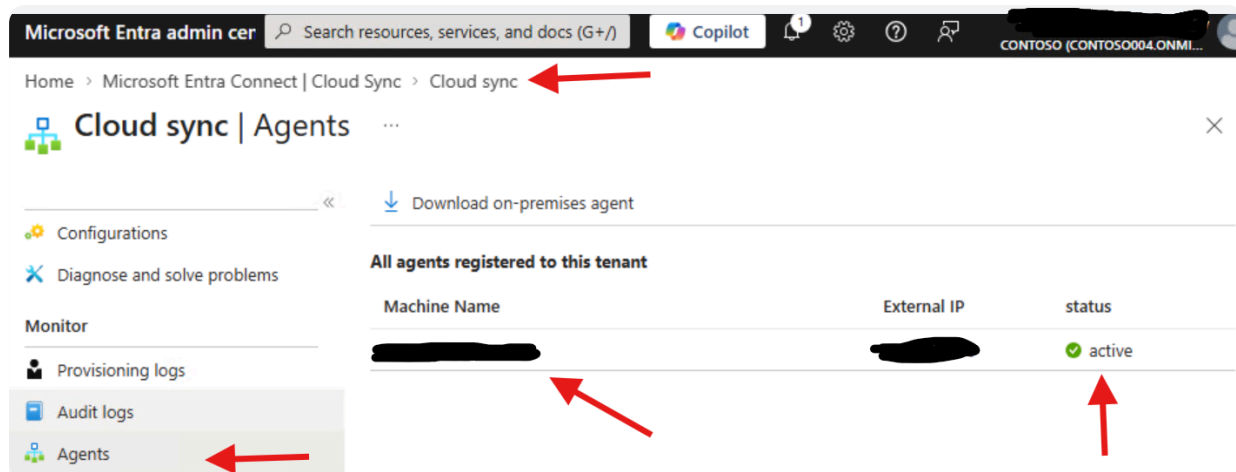
✓ WEVE MADE OUR FIRST CONNECTION FROM CLOUD TO ON-PREMISES

Microsoft ENTRA CLOUD SYNC



- Now Verify that the “Cloud Sync Agent has been installed in Entra from : “**Entra connect**” - **Cloud Sync** - “**Agents**” and “**All Registered Agents to This Tenant**”

That would be a YES



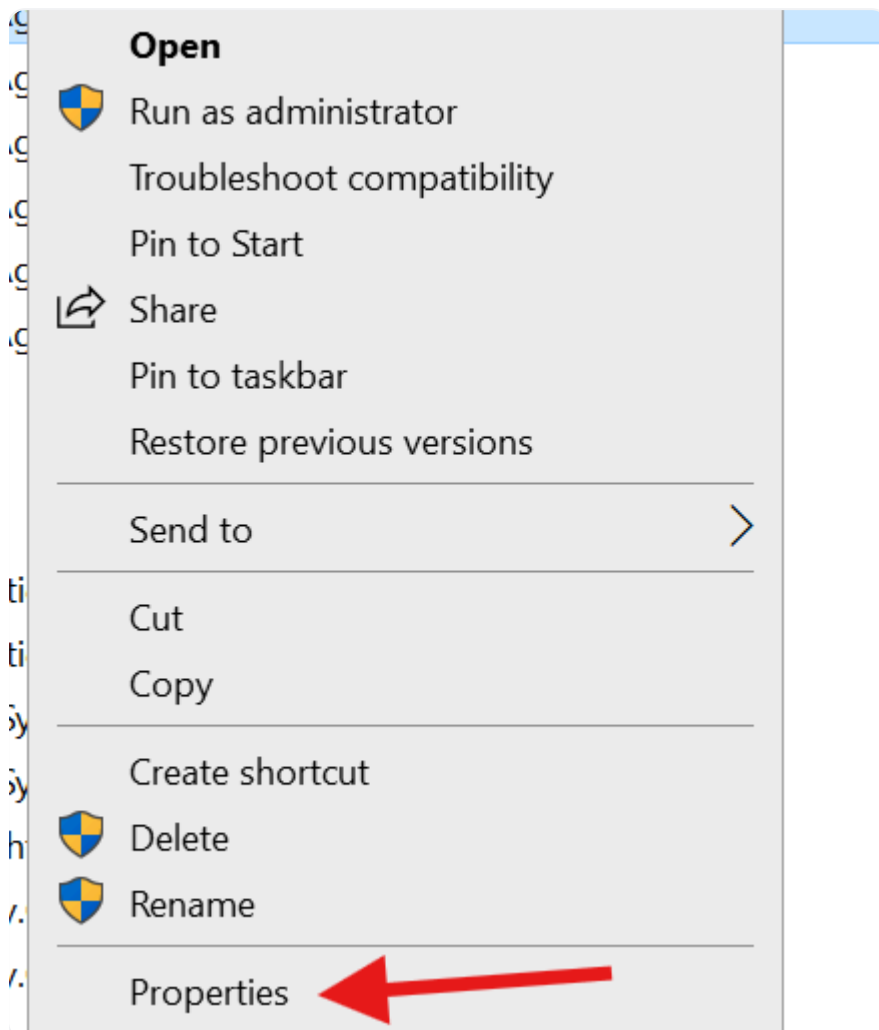
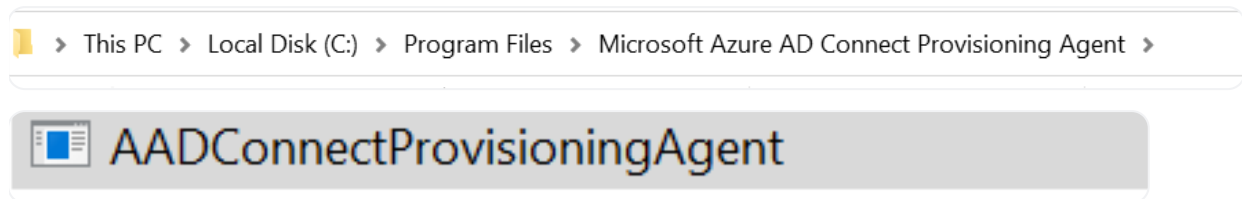
- And On your Windows SERVER Services “**Service.msc**” You will find the New “**AD Connect Provisioning Agent and Agent Updater**”

Service.msc

		Microsoft Azure AD Connect Agent Updater	Microsoft Az...	Automat
		Microsoft Azure AD Connect Provisioning Agent	A Windows ...	Automat

Verify your provision Agent version

1. Go to *C:\Program Files\Microsoft Azure AD Connect Provisioning Agent*.
2. Right-click *AADConnectProvisioningAgent.exe* and select Properties.
3. Select the Details tab. The version number appears next to the product version.



Product version 1.1.2108.0

GOOD TO GO!!!!

IMPORTANT: For the Next Step Adding “**New Configurations**” From On-Premises AD to Entra ID.

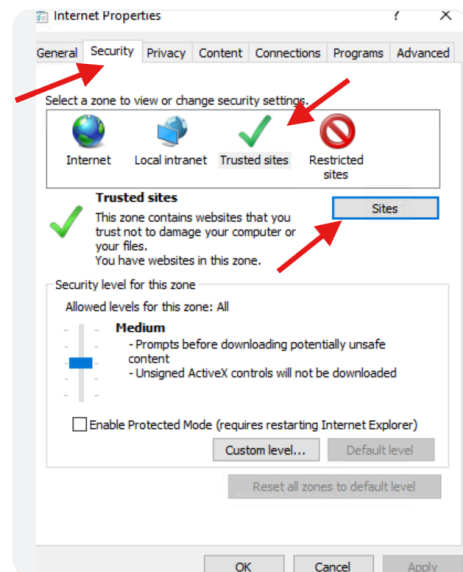
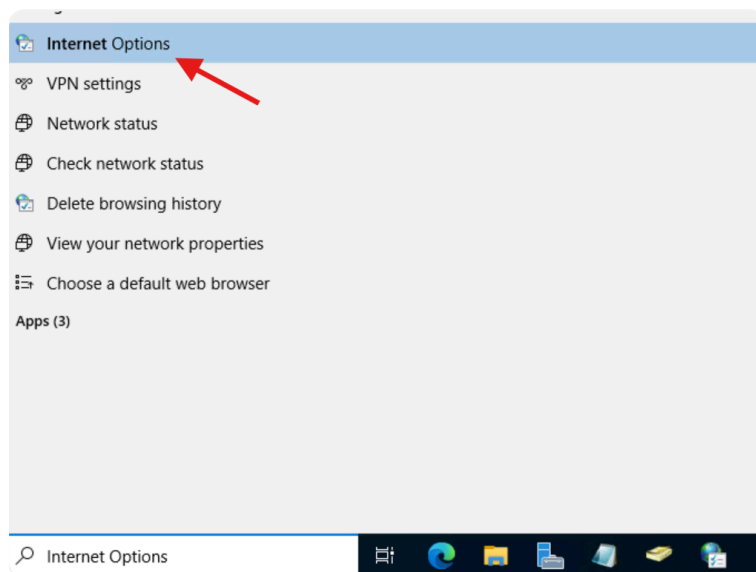
Be sure to “**Allow**” several needed addresses and ports within your On-Prem Server **BEFORE** continuing your Hybrid Configuration.

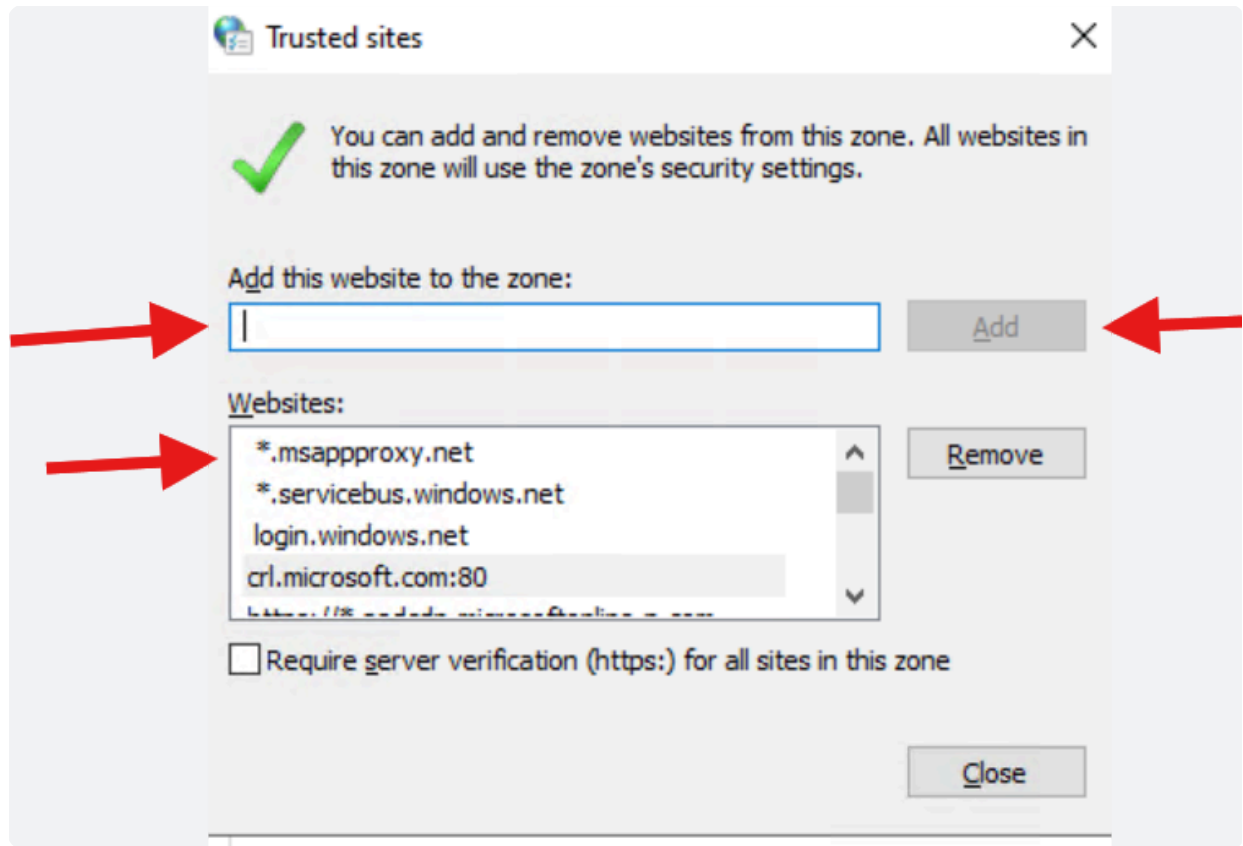
Within your On-Premises Server go to the Windows Searchbar and Type “**Internet options**”. Under “**Security**” add the following URL’s to your “**Trusted Sites Page**” in “**Internet Properties**”

-*.msaproxy.net. / .servicebus.windows.net

- login.windows.net / login.microsoftonline.com (These 2 for initial registration)

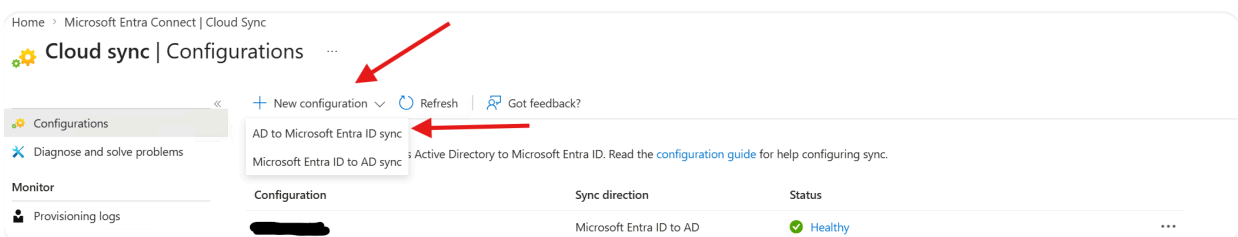
-mscrl.microsoft.com:80, crl.microsoft.com:80, ocsp.msocsp.com:80
www.microsoft.com:80





Once you've finished those URL additions go back to "**Entra Connect**" in the "**Entra Admin Center**" configure the Last of the "**Cloud Sync**"

- Now within **Entra Connect** from **Entra Admin center** - select "**Cloud Sync**",
Select the "**New-Configuration**" Drop down



- Select your Domain "**Enable Password hashing**" Then Click "**Create**"

New cloud sync configuration ...

Microsoft Entra ID

Create a Microsoft Entra Connect cloud sync configuration by completing the setup below with default parameters. You will be able to perform advanced configuration later in the setup. Read the [configuration guide](#) and [supported topologies](#) for setting up the configuration.

Which Active Directory domain would you like to sync? Note: We only show domains that have a reachable agent and have not been configured yet.

For a list of active agents, [click here](#).

☒ Enable password hash sync ⓘ

Next steps:

After creating your configuration with default parameters, you will be taken to the configuration details page to manage advanced settings.

Create

Cancel

Now we can see our cloud sync from both On-premises to Entra ID - and Entra ID to On-premises (AD) Status is **Healthy** and Ready to **Provision** its users

Cloud sync | Configurations ...

Configurations

Diagnose and solve problems

Monitor

Provisioning logs

Audit logs

Agents

Insights

Troubleshoot

New support request

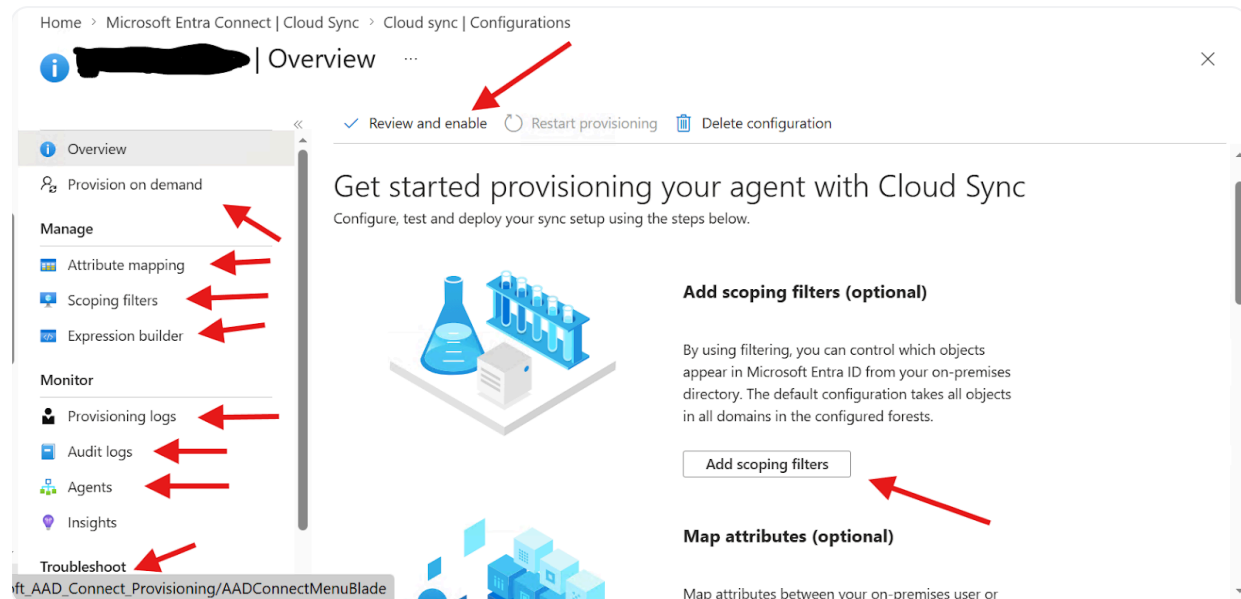
+ New configuration ▾ | Refresh | Got feedback?

Sync identities from on-premises Active Directory to Microsoft Entra ID. Read the [configuration guide](#) for help configuring sync.

Configuration	Sync direction	Status	
	Microsoft Entra ID to...	Healthy	...
	AD to Microsoft Entr...	Healthy	...

You will go back to the “**Entra Connect**” Page “**Cloud Sync**” And Go to the “**Cloud Sync Configuration Page**”

Go to "Review and Enable" Click and Finish setup



Now how do we know If we are Hybrid-Joined?

We Run a Test within the Command Line to check the Hybrid Status of Our **Active Directory Domain Controller "DC01"**

-Open Command-Line as Administrator and Type the Following command "**dsregcmd /status**"



Troubleshoot devices by using the dsregcmd command - Microsoft Entra ID

<https://learn.microsoft.com/en-us/entra/identity/devices/troubleshoot-device-dsregcmd>

```
Microsoft Windows [Version 10.0.20348.1487]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>dsregcmd /status
```

```
+-----+
| Device State |
+-----+

AzureAdJoined : YES
EnterpriseJoined : NO
DomainJoined : YES
DomainName : 
Device Name : Dc01
```

Device state

This section lists the device join state parameters. The criteria that are required for the device to be in various join states are listed in the following table:

[Expand table](#)

AzureAdJoined	EnterpriseJoined	DomainJoined	Device state
YES	NO	NO	Microsoft Entra joined
NO	NO	YES	Domain Joined
YES	NO	YES	Microsoft Entra hybrid joined
NO	YES	YES	On-premises DRS Joined

All devices ...

[Download devices](#) [Refresh](#) [Manage view](#) [Enable](#) [Disable](#) [Delete](#) [Manage](#) [Preview features](#)

[Add filters](#)

4 devices found

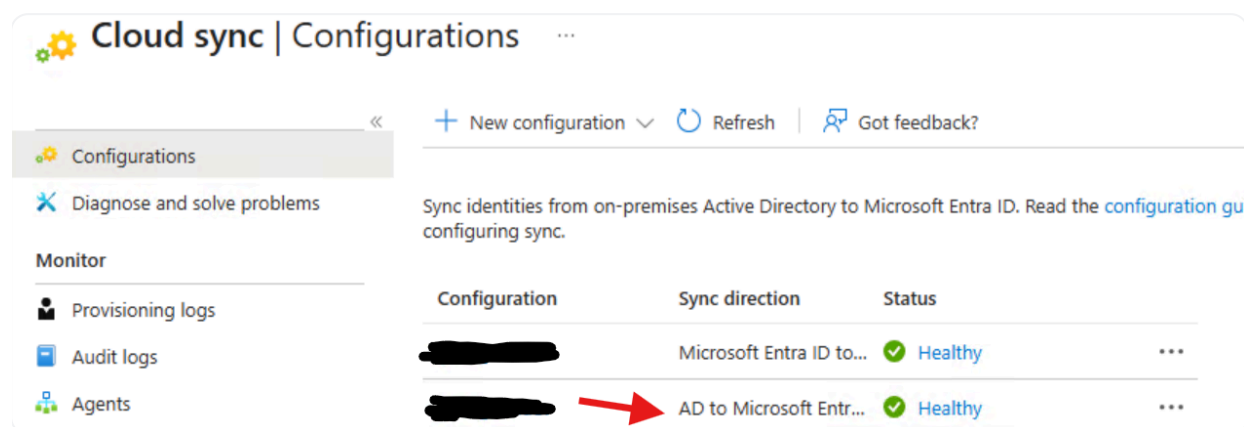
☐	Name ↑↓	Enabled	OS	Version	Join type
☐	 Dc01	 Yes	Windows	10.0.20348.1487	Microsoft Entra hybrid joined

✓ CONGRATS YOU ARE NOW HYBRID SYNCED!!!!

*For **Entra Connect Cloud Sync**, best practice for scoping is to use *Organizational Unit (OU)-based filtering to restrict synchronization to only necessary user and group objects. This approach provides a lightweight footprint, allows for granular control over hybrid identities, and enables using multiple agents for high availability.*

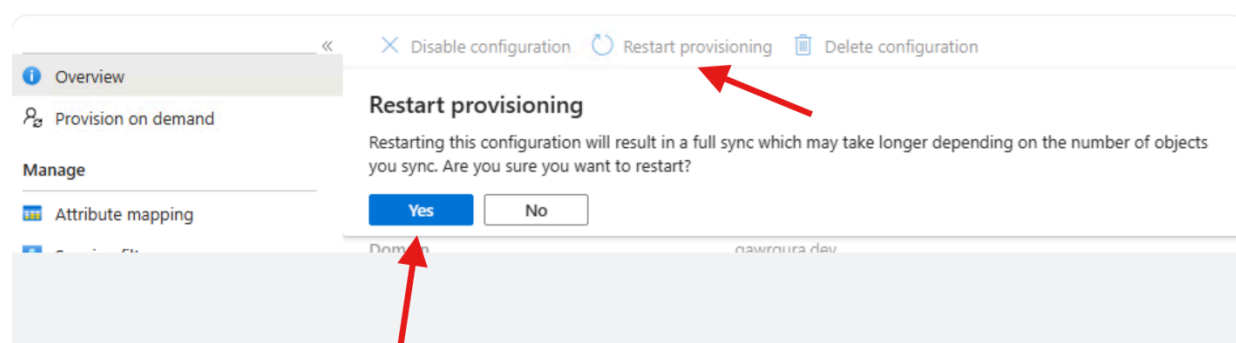
We Have a New User Named **"Jeremy"** In our **"ADUC"** Let perform a **"Provision Sync"** to Bring his User Object and Attributes to the Cloud

In the **"Cloud Sync"** Page go to the Connection with the **"Sync Direction"** **"AD to Microsoft Entra ID"**

The "Cloud sync | Configurations" page in the Microsoft Entra admin center. The left sidebar shows "Configurations" selected, with options for "Diagnose and solve problems", "Monitor" (including "Provisioning logs", "Audit logs", and "Agents"), and "Manage". The main content area shows a table of sync configurations. The table has columns for "Configuration", "Sync direction", and "Status". Two configurations are listed, both with a status of "Healthy". A red arrow points to the "AD to Microsoft Entr..." configuration.

Configuration	Sync direction	Status
[Redacted]	Microsoft Entra ID to...	Healthy
[Redacted]	AD to Microsoft Entr...	Healthy

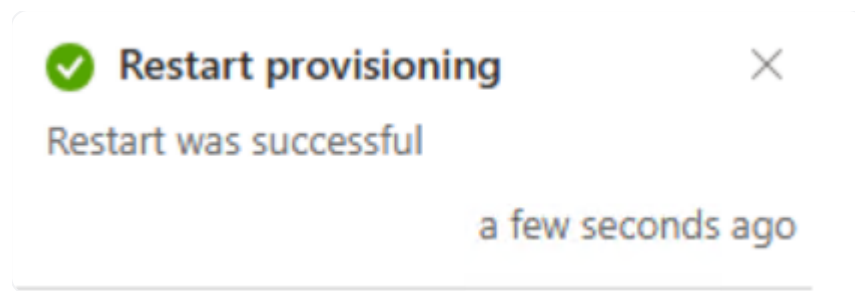
Now Lets **"Restart Provisioning"** to sync the user

A dialog box titled "Restart provisioning" with the text "Restarting this configuration will result in a full sync which may take longer depending on the number of objects you sync. Are you sure you want to restart?". There are two buttons: "Yes" (highlighted with a red arrow) and "No". Above the dialog, the "Restart provisioning" button in the configuration toolbar is also highlighted with a red arrow.

Restart provisioning

Restarting this configuration will result in a full sync which may take longer depending on the number of objects you sync. Are you sure you want to restart?

Yes No



Now we will find Jeremy within our "All Users" List within **Microsoft Entra Admin center** - **Users - All Users Scroll and Find his Name**

✓ CONGRATS YOU HAVE NOW HYBRID-SYNC YOUR USER FROM on-premises AD to M365 Entra Users

The screenshot shows the 'All Users' list in the Microsoft Entra Admin center. At the top, there are navigation links: 'New user', 'Edit', 'Delete', 'Download users (Preview)', 'Bulk operations', 'Refresh', and a menu icon. Below these is a search bar and an 'Add filter' button. A message states '19 users found'. The table has columns: 'Display name', 'User principal name', 'User type', 'Is Agent', 'On-premises sy...', 'Identities', and 'Company name'. The user 'Jeremy' is highlighted with a red arrow pointing to his row. Another red arrow points to the 'On-premises sy...' column for the same user. The user 'James' is also visible in the list.

Display name	User principal name	User type	Is Agent	On-premises sy...	Identities	Company name
HealthMailbox-EX01-010	HealthMailbox9/bt12d5e19...	Member	No	Yes	contoso004.onmicrosoft.com	
HealthMailbox-EX01-Mailbox-Dat	HealthMailbox952d7d8ff82...	Member	No	Yes	contoso004.onmicrosoft.com	
James	[REDACTED]	Member	No	No	contoso004.onmicrosoft.com	
Jeremy	Jeremy@[REDACTED]	Member	No	Yes	contoso004.onmicrosoft.com	
On-Premises Directory Synchroni...	ADToAADSyncServiceAccou...	Member	No	Yes	contoso004.onmicrosoft.com	

The user profile card for 'Jeremy' shows a red circular profile picture with a white 'J'. Below the picture is the name 'Jeremy' and the email 'Jeremy@[REDACTED]' with the role 'Member'. The card is divided into two columns of details. The left column includes 'User principal name', 'Object ID', 'Created date time', 'User type', and 'Identities'. The right column includes 'Group memberships', 'Applications', 'Assigned roles', and 'Assigned licenses'.

Basic info

 **Jeremy**
Jeremy@[REDACTED]
Member



User principal name	Jeremy@[REDACTED]	Group memberships	1
Object ID	0a701c0a-cd8d-45e2-b13b-a27861daf8d0	Applications	0
Created date time	Apr 6, 2026, 8:06 PM	Assigned roles	0
User type	Member	Assigned licenses	0
Identities	contoso004.onmicrosoft.com		

On-premises

On-premises sync enabled	Yes	
On-premises last sync date time	Apr 6, 2026, 8:15 PM	
On-premises distinguished name	CN=Jeremy,CN=Users,DC[REDACTED] 	
Extension attributes		
On-premises immutable ID	U1McFf8f+0GKozcJmk76hQ==	
On-premises provisioning errors		
On-premises SAM account name	Jeremy	
On-premises security identifier	S-1-5-21-1164286753-3891008217-2635403692-1165 	
On-premises user principal name	Jeremy@[REDACTED] 	
On-premises domain name	[REDACTED]	

Activate Windows
Go to Settings to activate Windows.

Additional Resources

<https://docs.azure.cn/en-us/entra/identity/hybrid/connect/how-to-connect-install-roadmap>

<https://learn.microsoft.com/en-us/exchange/hybrid-deployment/deploy-hybrid>

<https://learn.microsoft.com/en-us/exchange/hybrid-deployment-prerequisites>

<https://learn.microsoft.com/en-us/microsoft-365/admin/setup/add-domain>

<https://learn.microsoft.com/en-us/entra/identity/hybrid/connect/how-to-connect-install-prerequisites>



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